

SUJAY PATIL

☎ +44 07799271948 ✉ sujaypatil05327@gmail.com [in linkedin.com/in/sujay-patil-711429256/](https://www.linkedin.com/in/sujay-patil-711429256/) github.com/Sujay2478

Education

University of Glasgow

Integrated Master of Science (MSci) in Computer Science

Sep 2021 – July 2026

Glasgow, Scotland, UK

Experience

Glasgow University Software Service

Software Engineer

Mar 2024 – Mar 2026

Glasgow, Scotland, UK

- Translated Analogue Circuit Design Tuning MatLab Codes into Production Python Code using **ML algorithms**.
- Implemented **Machine Learning Techniques** to build and train **neural networks** for **constraint optimization**.
- Deployed **licensed proprietary software** compatible across multiple **Linux** distributions via a custom GUI installer.
- Built **test cases** to guarantee the optimisation algorithm's accuracy and efficiency.
- Tech Stack: **NumPy, PyTorch, MATLAB, GitLab**

Barclays

Technology Risk & Control Intern

Jun 2025 – Aug 2025

Glasgow, Scotland, UK

- Automated a **risk ownership mapping** process, reducing manual reporting effort by **2–3 hours per week**.
- Reduced outstanding **sub-limited JIRA tickets** by approximately **40%**, improving the team's risk remediation.
- Identified **40+ Cyber waiver tickets** to raise them to limited risk status in **ORAC** to enable tracking and oversight.
- Compiled an annual **supplier contractual obligation waiver** across **5 risk horizontals** to support risk compliance.
- Worked with technology, cyber, and risk teams to improve **governance processes, risk tracking, and remediation**.
- Enterprise Tools: **Atlassian Jira, ORAC, GitLab, Microsoft Excel, Confluence**

National Manufacturing Institute Scotland

Project Engineer Intern

Jun 2024 – Jun 2025

Paisley, Scotland, UK

- Developed an **AI-powered** system to extract, classify, and process invoice data for automated **carbon accounting**.
- Implemented complete **Scope 3 emissions** accounting in line with the **GHG Protocol**.
- Built a **Machine Data Analysis** system for **80+ machines** with real-time energy insights and automated reporting.
- Developed **interactive dashboards** to highlight load imbalance, idle consumption, and batch-level cost metrics.
- Worked directly with **engineers** and **stakeholders** to ensure the platform met business and technical requirements.
- Tech Stack: **Python, Django, React.js, SQL, OpenAI API, Figma, GitHub, Microsoft Excel, PowerBI**

GU Orbit

Head Software Engineer

Sep 2022 – Jun 2025

Glasgow, Scotland, UK

- Developed the prototype of the **processing pipeline** to be deployed on an **aerial testing platform**.
- Developed Python and C++ scripts and interfaced **BMP180 sensors** to **Raspberry Pi** with **i2c connection**.
- Successfully completed the **CloudView balloon launch**. The trip lasted 3 hours and sent back large amounts of data.
- Tech Stack: **C++, Python, Docker, Raspberry Pi, GitHub, Microsoft Teams**

Projects

AR-Assisted Sign Language Learning | *PyTorch, React.js, MediaPipe, ONNX Runtime Web*

Sep 2025 - Apr 2026

- For my Master's project, I researched **AR-assisted sign language education** by developing a browser based **BSL learning platform** with real-time gesture recognition and feedback. The system combined MediaPipe hand tracking, neural networks, and DTW based motion analysis to evaluate both static and dynamic gestures.

AI-Powered Life Cycle Assessment Tool - LCAi | *Django REST, React.js, OpenAI API*

Sep 2024 - Mar 2025

- In collaboration with the National Manufacturing Institute Scotland, I developed an AI-powered Lifecycle Assessment (LCA) tool designed to evaluate the **environmental impact** of products. The tool automatically analyzes **Bills of Materials (BOMs)** and related product data to assess environmental impacts across the product's entire lifecycle.

UofG Gaming Lab - Project Gundam | *Unity, C#, Google Cloud Platform (GCP)*

Sep 2023 - Mar 2024

- We developed a **Mobile AR application** for the **WorldCon 2024** event, using Unity. To encourage attendees, to explore Glasgow, this app aims to enhance visitors' experiences of the city. In addition the app also provides the user with a compass and a Gundam tracker, The app spawns several sorts of Gundam Models across Glasgow.

Technical Profile

Languages: Python, Java, C, C++, C#, HTML, CSS, JavaScript, TypeScript, SQL, MATLAB

Frameworks: Django REST, Flask, React.js, NumPy, Unity, PyTorch, Robot Operating System (ROS)

Skills & Tools: Git, Atlassian Jira, Confluence, Microsoft Excel, Power BI, Figma, Full-Stack Development, Artificial Intelligence, Machine Learning, Deep Learning Algorithms, Data Engineering, Data Analysis, Business Analytics, Cloud Computing, Risk Analysis, Solutions Engineering, Project Management, Database Modelling